was comfortable with that. Infinity sounds like a big number, but we use infinite quantities in physics all the time. The number of real numbers between 0 and 1 is infinite, as you know. If Hilbert space is infinite-dimensional, it doesn't make much sense to talk about the number of individual worlds. But we can group a set of similar worlds together, and talk about the total weight (amplitude-squared) they have compared to some other group."

"Great. So Hilbert space is infinite-dimensional, and the number of worlds is infinite, but you want to claim that we should only talk about the relative weight of different kinds of worlds?"

"No, I'm not done yet," Alice insisted. "The real world isn't a bunch of particles, nor is it even described by quantum field theory."

"It's not?" said her father in mock dismay. "What have I been doing all my life?"

"You've been ignoring gravity," replied Alice, "which is a perfectly sensible thing to do while you're thinking about particle physics. But there are indications from quantum gravity that the number of distinct possible quantum states is finite, not infinite. If that's true, there is a maximum number of worlds we could sensibly talk about, given by the dimensionality of Hilbert space. The kinds of estimates that get thrown around for the number of dimensions of the Hilbert space of our observable universe are things like $2^{10^{122}}$. A big number," Alice admitted, "but even very big finite numbers are much smaller than infinity."

Her father seemed to think about this. "Huh. I'm not really sure we know anything very reliable about quantum gravity—"

"Maybe we don't. That's why I said we really don't know if the number of worlds is finite or infinite."

"Fair enough. But that raises a totally new worry. It seems to me that branching should be happening all the time, every time a quantum system becomes entangled with its environment. Is it conceivable that this number you just quoted, while mind-bogglingly large, isn't large enough? Are we sure there's enough room in Hilbert space for all the branches of the wave function that are being produced as the universe evolves?"

"Hmm, I never thought about that, to be honest." Alice grabbed a napkin and started scribbling some numbers on it. "Let's see, there are about 10^{88} particles within our observable universe, mostly photons and neutrinos. For the most part these particles travel peacefully through space,